

Meta-learning In-Context Enables Training-Free Cross Subject Brain Decoding

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Abstract

*Visual decoding from brain signals is a key challenge at the intersection of computer vision and neuroscience, requiring methods that bridge neural representations and computational models of vision. A field-wide goal is to achieve generalizable, cross-subject models. A major obstacle towards this goal is the substantial variability in neural representations across individuals, which has so far required training bespoke models or fine-tuning separately for each subject. To address this challenge, we introduce a meta-optimized approach for semantic visual decoding from fMRI that **generalizes to novel subjects without any fine-tuning**. By simply conditioning on a small set of image-brain activation examples from the new individual, our model rapidly infers their unique neural encoding patterns to facilitate robust and efficient visual decoding. Our approach is explicitly optimized for in-context learning of the new subject's encoding model and performs decoding by hierarchical inference, inverting the encoder. First, for multiple brain regions, we estimate the per-voxel visual response encoder parameters by constructing a context over multiple stimuli and responses. Second, we construct a context consisting of encoder parameters and response values over multiple voxels to perform aggregated functional inversion. We demonstrate strong cross-subject and cross-scanner generalization across diverse visual backbones without retraining or fine-tuning. Moreover, our approach requires neither anatomical alignment nor stimulus overlap. This work is a critical step towards a generalizable foundation model for non-invasive brain decoding. Code and models are publicly available at <https://github.com/ezacngm/braInCodec>.*

1. Introduction

Developing robust theories of intelligence requires generalizable, population-wide models of human brain function. An important step has been the development of high-fidelity visual decoders of brain activity [15, 99], enabled by conditional image generation models and the availability of high-quality fMRI visual datasets. Visual reconstruction serves as a unique and demanding testbed for conditional generation, requiring vision models to synthesize images from signals that are not only noisy but also highly abstract. A common strategy decomposes this challenge into two sub-problems: (1) learning a mapping from high-dimensional brain activity to a compact visual-semantic representation; and (2) synthesizing naturalistic images from that representation. The synthesis challenge has been addressed by leveraging large-scale generative models as image priors [85, 88]. Simultaneously, high-quality neural activity datasets [2, 14, 35, 44, 45, 59] at scale have provided sufficient data to solve the mapping sub-problem on an individual basis, driving the recent surge in high-fidelity, within-subject reconstructions.

Despite this recent progress, a critical barrier prevents widespread application of brain decoding: current models **cannot generalize across subjects**, necessitating per-subject models or subject-specific fine-tuning [94, 106]. This challenge is rooted in the profound inter-subject variability in neural signals which arises from complex interacting sources [101], including differences in anatomical structure and functional organization shaped by development, individual experience, and neuroplasticity [12, 32, 100, 107]. As a result, the mapping function learned for one individual is ineffective for another, necessitating retraining or fine-tuning via gradient descent, a data-intensive and computationally demanding process. Developing a data-efficient

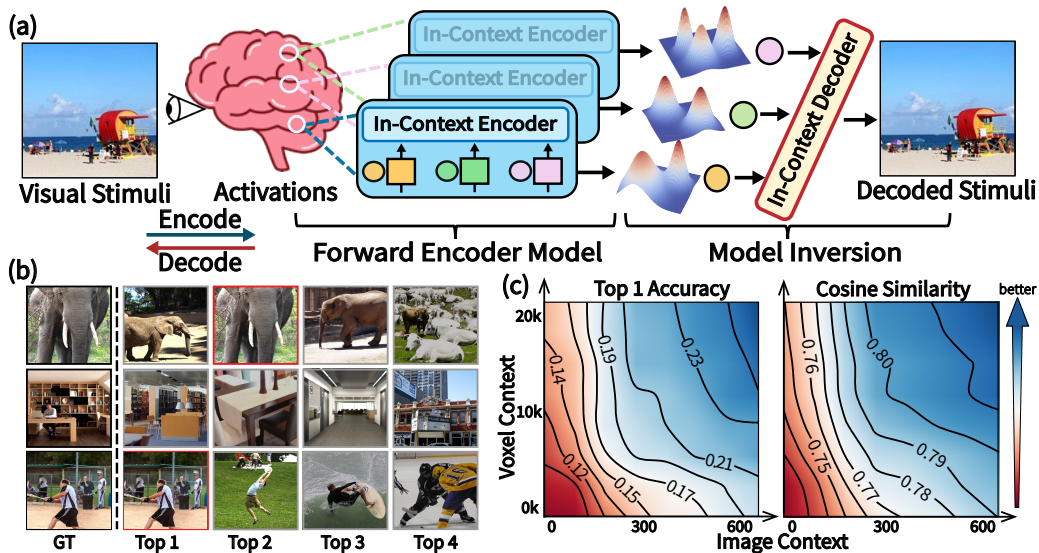


Figure 1. **Overview of our hierarchical brain decoding framework.** Encoders predict brain activity from stimulus, while decoders reconstruct stimulus from brain activity. (a) Our framework can generalize to novel subjects **without any fine-tuning**. In the first stage, we infer parameters of a forward model (image-computable encoder) by constructing a context using stimuli/activity pairs for a single voxel, repeated for every voxel. In the second stage, we construct a context across multiple voxels, fusing the encoder parameters with observed brain activations to decode the stimuli. Our approach requires neither anatomical alignment, nor stimulus overlap. (b) Decoding results on BOLD5000 after training on NSD, our method can **generalize without fine-tuning across scanners, voxel size, and subjects**. (c) Our model performance positively scales with both the number of images provided in Stage 1, and the number of voxels provided in Stage 2.

generalizable cross-subject visual decoding model is therefore essential for building population-wide theories and for enabling applications in brain-computer interfaces (BCIs), cognitive assessment, and personalized diagnostics.

A principled approach is to recognize that neural decoding is fundamentally an inverse problem. A robust solution should be constrained by an accurate forward model of the system that characterizes how the brain of an individual subject represents information. In computational neuroscience, this forward model is referred to as an “encoding model” [73], which predicts brain activity from stimuli. Meanwhile, the inverse operation is performed by the decoding model. Following this principle, our approach structures the decoding process as a functional inversion problem that we solve hierarchically. **First**, we estimate the visual response function weights for individual voxels in-context [114]; **Second**, we build a decoder that performs contextual integration across multiple brain regions to perform a subject-specific functional inversion to reconstruct the visual stimulus. This two-stage in-context learning process enables generalization to novel subjects **without any fine-tuning** and **with relatively small amounts of new data**. Since image synthesis from brain activity has been well explored using pretrained generative models, we instead focus on decoding image embeddings from novel subjects.

We name our method BrainCoDec (Brain In-Context Decoding), and outline the approach in Figure 1. Concretely: (1) Our method generalizes to novel subjects, requires no

anatomical alignment or stimulus overlap, and is the first to work across different scanners and acquisition protocols without gradient-based finetuning. (2) Through selective dropout of functionally specialized regions and by using only a small subset of voxels from higher visual cortex, we demonstrate strong robustness to input variability. (3) Attention visualizations across images from diverse categories reveal interpretable spatial maps that align closely with known functional regions of the visual cortex. This approach marks a significant step towards a truly universal and scalable brain foundation model for investigating neural representations across the human population.

2. Related work

Computational Encoding and Decoding Models. Computational analysis of neural data usually leverage two complementary approaches. Encoding models predict neural activity from stimuli, and decoding models that reconstruct stimuli from brain activity [23, 34, 40, 48, 67, 73, 76, 87, 95, 96]. Both approaches have benefited from the development from feature extractors trained on large-scale datasets, with the dominant approach leveraging linear mappings from learned features to neural activity [25, 27, 33, 38, 52, 93], with more recent approaches utilizing attention based parameterization [1, 4, 6]. Core to our current work is the approach proposed in [114], which meta-optimizes an encoding model to generalize to novel subjects. Encoders can be used to investigate the selectivity in visual cortex [26, 50, 51, 57, 65, 90, 110, 111], or combined with

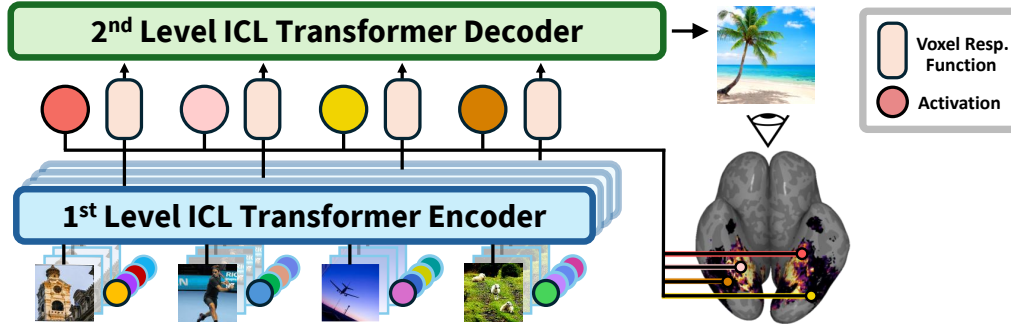


Figure 2. **Model architecture of BrainCoDec.** In stage one, the in-context encoder infers encoder parameters by in-context learning across stimuli/activation pairs for a single voxel. This is repeated across the voxels of interest. In stage two, we integrate across multiple voxels, taking as input the voxelwise parameters and activation corresponding to a novel image. Both stages can vary the context sizes.

generative models to synthesize new stimuli [5, 13, 37, 63, 64, 69, 81, 82, 86, 104]. By leveraging generative models, stimulus can be decoded from fMRI, EEG, and MEG for images [8, 15, 24, 28, 39, 58, 61, 62, 66, 68, 79, 94, 99], dynamic visual stimuli [16, 30, 36, 60, 91, 113, 115], and speech/audio/language [7, 47, 70, 78, 80, 102, 108]. Recent work seeks to achieve generalization via flatmaps [56, 105], 1D pooling [106] or surface learning [21], these approaches implicitly (flatmaps & pooling) or explicitly (surface) require anatomical alignment.

Inverting Encoding Models for Decoding. Prior work has sought to decode (identify the category or semantic nearest neighbor) of stimuli by comparing patterns of neural activations [42, 43, 53, 77]. Reconstructing viewed images from neural activity by inverting a forward model (encoder) has been previously demonstrated using simple stimuli [9], which inverts the encoder using ordinary least squares to solve for the color of the image. Similar approaches that convert between encoders that predict neural activation and decoders have been utilized in the context of motion direction [54, 89], orientation [10], and more complex stimuli like faces [20, 41], natural images [49, 72, 92] and movies [75]. Generally these methods are based on the principle of matching stimuli and their predicted brain activity to the true observed brain activity, and decoding the stimuli by solving or identifying the solution. Our learned approach significantly extends this prior work by functioning even when the system is under-determined (fewer voxels than stimulus representation), and being able to account for biases in the encoder estimation.

Meta-Learning and In-Context Learning. Meta-learning focuses on training models to rapidly adapt to new tasks by leveraging prior knowledge acquired from a distribution of related tasks [46]. It facilitates fast generalization to novel problems with few examples and minimal training effort. Classic approaches include meta-optimization methods [29, 74, 84] and metric-based formulations [97]. In parallel, large language models display strong in-context learning (ICL) capability [11, 103]: given prompts with

demonstrations, model behaviors could be adjusted at inference time effectively without updating parameters [18, 71]. These observations may suggest that in-context learning serves as an implicit meta-learning mechanism, whereby transformers develop internal adaptation procedures during the pretraining stage [22, 31]. In our work, which aims to learn the functional mapping between visual stimuli and voxelwise brain responses, we construct a framework that integrates meta-training with in-context learning. This approach enables training-free adaptation to novel subjects.

3. Methods

Our method is based on the learned inversion of a set of encoders. The framework leverages meta-learning, and uses few-shot, in-context examples for the decoding of unseen stimuli (Figure 2). For unseen subjects, this approach does not require **any fine-tuning**. We first define the problem in Section 3.1, and discuss how stimuli can be recovered by inverting a set of encoders in Section 3.2. In Section 3.3 we discuss how hierarchical in-context learning can enable training-free decoding on novel subjects. Since image generation is relatively well studied, in this work we focus on decoding an image embedding, as it is core to the mapping problem, and evaluate method performance using retrieval following [55, 94, 109].

3.1. Motivation and Problem Definition

Substantial cross-subject variability in neural responses poses a major obstacle to generalizable brain decoding. Rather than directly learning a fixed inversion mapping, we reformulate neural decoding as a meta-learning problem that learns **how to perform functional inversion**. Crucially, our approach does not rely on any shared stimuli or anatomical alignment across subjects.

Formally, let an image I be represented by its embedding vector $\mathcal{I} = \phi(I) \in \mathbb{R}^{1 \times d}$, where ϕ denotes a pre-trained image feature extractor such as CLIP [83], and d is the embedding dimension. For a given image stimulus I ,

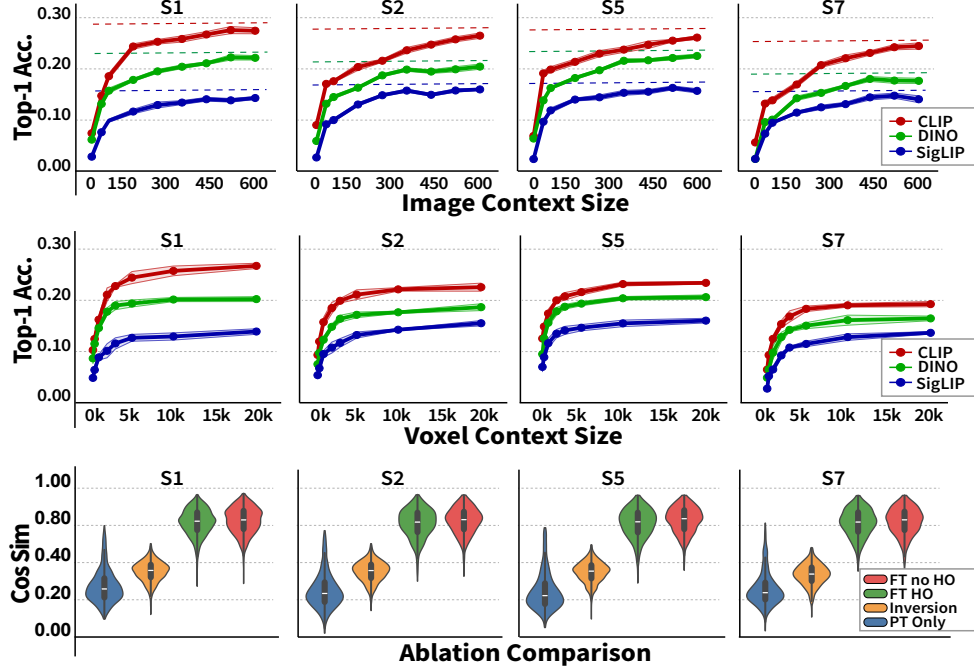


Figure 3. **Contextual scaling and ablation analysis of BrainCoDec.** **Top:** Image-context scaling from stage 1. Decoder performance scales positively with more images collected for the novel subject. **Middle:** Voxel-context scaling from stage 2. Top-1 retrieval accuracy improves consistently as the number of in-context voxels increases with all visual backbones across all subjects. **Bottom:** Ablation comparison. Cosine similarities for four variants using CLIP backbone, synthetic data pretraining (PT Only), gradient inversion (Inversion), training with subject hold-out (FT HO; BrainCoDec), and training on seen subjects (FT no HO). These results show that models trained with real neural data outperform the models trained with only synthetic data, with only marginal gains from fine-tuning on a subject.

the corresponding fMRI response for a subject is denoted as $B_{\mathcal{I}} = (\beta_1, \beta_2, \dots, \beta_K)_{\mathcal{I}} \in \mathbb{R}^{1 \times K}$, where K is the number of voxels in the subject’s visual cortex. During testing, for a new subject we observe a small set of n context image-brain activation pairs $\{(\mathcal{I}_i, B'_i)\}_{i=1}^n$, where B'_i represents the measured voxel activations for the i -th image. Our goal is to infer the embedding $\mathcal{I}_{\text{novel}}$ of an unseen image from its corresponding brain response B'_{novel} using only these context examples.

3.2. Decoding as the Functional Inversion

Let us assume that the forward model (image-computable encoder) predicts for a given voxel v_k : $f_k(\mathcal{I}) \Rightarrow \hat{\beta}_{\mathcal{I},k}$. Ideally, given a sufficient number of voxels $\{v_1, v_2, v_3, \dots, v_j\}$ where $j \gg d$ and encoder functions that are error free, we can uniquely solve for the stimulus $\mathcal{I}^*$ by inverting the encoding model such that:

$$\mathcal{I}^* = \arg \min_{\mathcal{I}} \left(\sum_m^j \|f_m(\mathcal{I}) - \beta_m\|_2^2 \right) \quad (1)$$

In practice, the forward models of the encoders could be biased and inaccurate, the choice of metric/distance may affect the solution, and knowledge about the distribution of the inputs or outputs may improve the decoder. Unlike prior work that learn decoders to map from neural representations

to stimuli directly, our approach takes a meta-learning view and learns a model to perform in-context functional inversion across a variable number of higher visual cortex voxels.

3.3. Hierarchical Training-Free Stimulus Decoding

Our decoding approach leverages a hierarchical inference process, with two successive in-context stages, each with a distinct type of context. In **Stage 1**, we perform in-context inference across multiple stimulus-response pairs to infer the voxelwise response (encoder) function parameters. We run this per-voxel, across all voxels of interest. In **Stage 2**, we construct a voxel context across multiple voxels to perform inversion and estimate the image embedding. Here the context consists of an aggregate of voxelwise encoder parameters and activations for a single novel stimulus.

Encoder Parameter Estimation. In Stage 1 we adopt BrainCoRL’s approach [114] to estimate the per-voxel parameters. For a novel subject, for voxel v_q we have a context defined by $\{(\mathcal{I}_1, \beta_{1,q}), (\mathcal{I}_2, \beta_{2,q}), \dots, (\mathcal{I}_n, \beta_{n,q})\}$, where we have the voxel’s activation in response to n images. Let the pretrained BrainCoRL model be T_θ , then:

$$\omega_q = T_\theta(\{(\mathcal{I}_t, \beta_{t,q})\}_{t=1}^n) \quad (2)$$

where the model can output the voxelwise function weights of a novel subject without any fine-tuning. Note that we per-



Figure 4. **Image retrieval comparison on a subject unseen during training (S1).** For each method (BrainCoDec-200, MindEye2 + anatomical alignment, TGBD), columns list the Top-1–3 retrieved images out of 907 test images from left to right, ranked by similarity in the evaluation embedding space. Red boxes mark correct hits. Our model can yield very high semantic retrieval consistency without any fine-tuning.

Table 1. **Quantitative comparison of unseen subject brain decoding performance.** Top-1 and Top-5 retrieval accuracy (%) on unseen NSD subjects (S1, S2, S5, S7) for MindEye2 + anatomical alignment, TGBD, and BrainCoDec-200 (200 in-context images). Our method substantially outperforms prior methods while requiring neither subject-specific fine-tuning nor large-scale training data. Mean accuracies across subjects are reported in the rightmost column; additional metrics and standard deviations are provided in the **Appendix**.

Models	S1		S2		S5		S7		Mean	
	Top-1↑	Top-5↑	Top-1↑	Top-5↑	Top-1↑	Top-5↑	Top-1↑	Top-5↑	Top-1↑	Top-5↑
MindEye2 [94]	4.11%	12.9%	3.82%	10.70%	2.87%	9.58%	2.51%	6.49%	3.90%	9.81%
TGBD [55]	1.27%	3.89%	0.56%	2.33%	0.84%	3.34%	0.39%	1.41%	0.82%	3.09%
BrainCoDec-200	25.5%	56.6%	22.9%	52.4%	23.2%	55.8%	19.2%	51.2%	22.7%	54.0%

form this stage independently for each voxel in higher visual cortex, computing contextual structure across stimuli separately for each voxel.

Contextual Functional Inversion. In Stage 2, the model performs functional inversion by constructing a context across voxels within a single subject. This approach allows us to flexibly adapt our model to novel subjects which have different voxel counts. Our approach does not require any reference to anatomy, and does not require cross-subject anatomical alignment. Each voxel v_k is represented by a context token c_k , defined as the concatenation of its predicted response parameter ω_k derived from stage 1, and the measured activation β_k from the novel stimulus, $c_k = [\omega_k, \beta_k]$. The voxel context for a subject is then $\{c_k\}_{k=1}^m$, where $m \leq K$. We train a transformer P_γ with variable-length voxel contexts to approximate the aggregated inverse mapping:

$$\hat{\mathcal{I}} \approx P_\gamma(\{c_k\}_{k=1}^m) \quad (3)$$

where P_γ denotes a learned transformer that jointly inverts the functional representations of multiple voxels.

Test-time Context Scaling. At test time, when a new subject is presented, the number of K voxels available for decoding may vary across individuals. This variability in context size poses a challenge for model generalization. Unlike transformers in language modeling, where outputs depend on the sequential order of tokens, our model should be invariant to both the number and the order of voxel token inputs. To accommodate variable-length contexts, we adopt logit scaling [3, 17, 98]. Assuming a query/key (q, k) with d features and a length l context:

$$\alpha_{\text{orig}} = \frac{q \cdot k}{\sqrt{d}}; \quad \alpha_{\text{scaled}} = \frac{\log(l) \cdot q \cdot k}{\sqrt{d}} \quad (4)$$

Our model integrates a [CLS] token for output. We omit positional embeddings to achieve order invariance.

Training Objective. To achieve both fine-grained alignment and instance-level discriminability, we employ a hybrid cosine-contrastive loss that combines cosine embedding loss and an InfoNCE loss. Let $\mathcal{I}$ be unit vectors:

$$\mathcal{L}_{\text{total}} = (\mathcal{L}_{\text{cos}} + \alpha \mathcal{L}_{\text{InfoNCE}}) \quad (5)$$

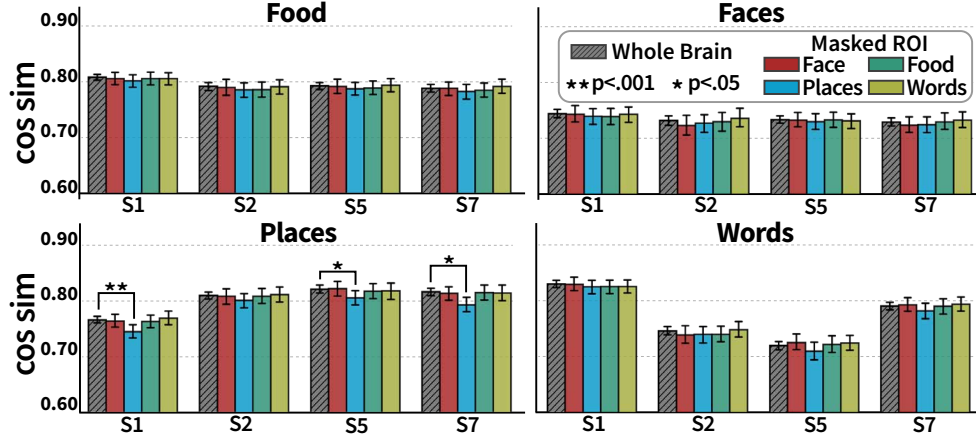


Figure 5. **Robustness of removing voxels from ROIs.** Cosine similarity of masking out category-specific voxels (Food, Faces, Places, Words) across four unseen NSD subjects on top-activating images from the test set. For each category, we compare performance using full context voxels from higher visual cortex versus masking out category-selective ROIs. Across nearly all conditions, masking the corresponding functional region has minimal impact on decoding performance, indicating strong robustness and distributed representation learning in BrainCoDec. Masking scene-selective regions (PPA/OPA/RSC) leads to some performance drop.

where for a batch size of N :

$$\mathcal{L}_{\text{total}} = \frac{1}{N} \sum_{i=1}^N \times \left((1 - \hat{\mathcal{I}}_i^T \mathcal{I}_i) - \log \frac{\exp(\hat{\mathcal{I}}_i^T \mathcal{I}_i / \tau)}{\sum_{j=1}^N \exp(\hat{\mathcal{I}}_i^T \mathcal{I}_j / \tau)} \right)$$

We found this loss to work well for our task, as it optimizes both reconstruction and discriminability.

4. Experiments and Analysis

In this section we comprehensively evaluate BrainCoDec’s capability. We first describe the experimental setup in Section 4.1. We then examine the effectiveness on unseen subject generalization in Section 4.2, and the decoding robustness in Section 4.3. Next, we investigate the model’s internal representational structure via attention-based analyses in Section 4.4. Finally, we evaluate its ability to adapt to new scanner, voxel sizes, and scanning protocols on the BOLD5000 data in Section 4.5. Together, these experiments provide a rigorous characterization of the model’s decoding capability, robustness, and interpretability.

4.1. Experiment Setup

Dataset. We evaluate model performance on Natural Scenes Dataset (NSD) [2] and further validate on BOLD5000 [14]. Both are large-scale fMRI datasets. NSD is the largest available 7T neural dataset, in which each subject viewed $\sim 10,000$ images for up to three times. There is no overlap between train and test images. BOLD5000 is a 3T dataset, in which each subject viewed $\sim 5,000$ images, but only a subset of images was viewed four times.

For NSD, four subjects (S1, S2, S5, S7) completed the whole scanning among all eight subjects, and thus are mainly used in our experiments. For each NSD subject, roughly 9,000 images are uniquely seen by to that subject, while $\sim 1,000$ images are commonly viewed by all eight

subjects. To rigorously evaluate BrainCoDec on novel subjects, we use the $3 \times 9,000$ unique images from three subjects as meta-training data, the $1 \times 9,000$ unique images from one held-out subject as the support image context, and the 1,000 common images viewed by the held-out subject as the final test set. We perform analyses in subject-native volume space (func1pt8mm) for all NSD subjects. For the data preprocessing, voxelwise betas are z -scored within each session and then averaged across repeats of the same stimulus. For ROI-level evaluations, we apply a t -statistic threshold of $t > 2$ using independent functional localizer data provided with the dataset to refine broad ROI definitions following prior work [64]. For quantitative evaluations, we apply a voxel-quality cutoff of $\text{ncsnr} > 0.2$ following [19]. For BOLD5000, we use a model trained on the four NSD subjects (no subject held-out) and evaluate directly on BOLD5000 subjects (CSI1, CSI2, CSI3) without additional training using 5-fold cross-validation. We only utilize those stimuli with four repeats and apply a cutoff of $\text{ncsnr} > 0.3$ as the dataset authors recommend. Voxel stimuli responses are averaged over all the repeats.

Training Strategy. Our training strategy is inspired by LLM pipelines and consists of three stages: pretraining, contextual extension, and supervised fine-tuning. In the pretraining stage, we adopt an analysis-by-synthesis scheme that does not use any real fMRI data. We simulate a large population of voxels by sampling synthetic weights and corresponding beta responses with random Gaussian noise, and train the model with a fixed voxel-context size of 200. In the second stage, we introduce variable-length contexts by randomly drawing the number of voxels from Uniform(200, 4000), enabling the model to become robust to changes in context length. In the final fine-tuning stage, the model is optimized on real fMRI measurements, us-

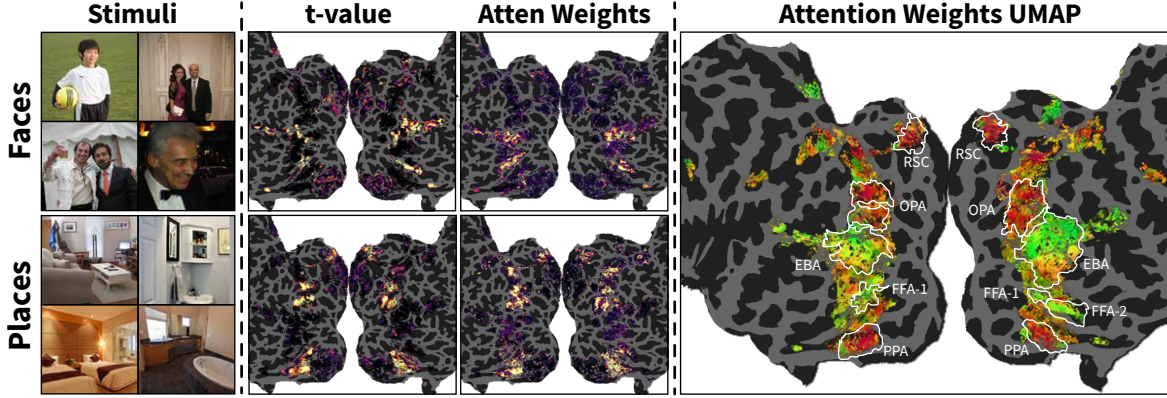


Figure 6. **Semantic attention patterns in BrainCoDec.** **Left:** Example face and place stimuli used for category-specific analysis. **Middle:** Comparison of category t -values from an independent NSD functional localizer, with the corresponding attention-weight maps from the final self-attention layer when decoding these stimuli, showing closely matched spatial distributions. **Right:** UMAP projection of voxelwise attention weights across the full test set. Color-coded clusters separate body/face-selective regions in green (EBA, FFA/aTL-faces) and scene-selective regions in red (RSC, OPA, PPA).

ing subject-specific beta values and voxel response parameters estimated by the pretrained BraInCoRL across different image-context sizes, leading to fast convergence and effective adaptation to biologically realistic neural signals.

Evaluation Metrics. We evaluate cross-subject decoding on the foundational nearest-neighbor image retrieval task, which accurately reflects the capabilities of decoding models. Our method can also be extended to reconstruction tasks by incorporating an additional pretrained image generator such as IP-Adapter [112] and Stable Diffusion [88]. To quantitatively compare with other methods, we adopt 4 decoding quality evaluation metrics following [55, 94], top-1 accuracy, top-5 accuracy, mean rank, and cosine similarity. To note **all our evaluation experiments are performed on novel subjects that are unseen by the model during training**, with exception of the no subject held-out (“no HO”) in the ablation study in Figure 3.

4.2. Unseen Subject Brain Decoding

SOTA Method Comparison. We evaluate unseen subject decoding image retrieval task with CLIP backbone following the MindEye2 protocol [94]. We compare against two state-of-the-art methods, MindEye2 [94] and TGBD [55], across all four leave-one-subject-out conditions. For fair comparisons, TGBD is retrained using its official recipe using the same dataset split as ours; MindEye2 is evaluated using its official released fine-tuned model with MNI volume anatomical alignment when inferring on novel subjects. We report the limited-context variant **using only 200 of the 9000 support images**, denoted BrainCoDec-200. Quantitative and qualitative results appear in Table 1 and Figure 4, respectively. As shown above, BrainCoDec delivers consistently stronger retrieval performance than both baselines on the generalizations to unseen subjects without retraining.

Contextual Scaling. We investigate how BrainCoDec’s performance scales with the two aspects of context, image context and voxel context, respectively. The results are shown in Figure 3. A clear scaling pattern emerges across all subjects and visual backbones (CLIP, DINO, and SigLIP). Increasing either the image or voxel context size consistently improves decoding. Remarkably, with only 200 images and 4,000 voxels, BrainCoDec achieves similar accuracy as inference using full context (all $\sim 9,000$ images and all higher-visual-cortex voxels). This shows that our framework requires only a fraction of subject-specific data to reach comparable decoding performance.

Ablation Study. We compare four configurations, BrainCoDec with synthetic data pretraining only, gradient-based functional inversion, BrainCoDec trained with real data with or without subject holdout (seen subject scenario). As illustrated in Figure 3, both fine-tuned variants significantly outperform the pretrained-only and direct-inversion baselines, confirming the effectiveness of BrainCoDec. The performance gap due to subject holdout is marginal. In contrast, models trained with pretraining only or direct inversion exhibit substantially lower cosine similarity, underscoring the necessity of contextual fine-tuning for accurate cross-subject decoding.

4.3. Robust Decoding through ROI Dropout

We examine if BrainCoDec **requires** functionally specialized cortical regions during decoding. For each semantic category (faces, places, food, and words), we first identify the test images that elicit the strongest mean beta activations within the corresponding functional voxels. We then systematically mask out the corresponding category-selective regions (e.g., removing PPA, occipital place area (OPA), and retrosplenial cortex (RSC) for scene-related stimuli)

and evaluate the resulting decoding performance. As shown in Figure 5, the model exhibits remarkable robustness to such targeted regional dropout. Masking category-related ROIs leads to minimal degradation for most categories, indicating that BrainCoDec does not rely on any single functional region to perform aggregated decoding.

4.4. Neural Interpretability via Attention Analysis

We analyze the internal attention dynamics of BrainCoDec by extracting the attention weights from the last layer during the decoding of test images belonging to distinct semantic categories using the same activation-based selection criterion as before. As visualized in Figure 6, the learned attention weights reveal highly interpretable spatial patterns. Face-related stimuli elicit elevated attention weights in voxels in the face- (FFA) and body-selective (EBA) regions, while place-related stimuli elicit elevated attention weights in place-related regions (PPA, OPA, and RSC). These results confirm that BrainCoDec learns to allocate selective focus consistent with established cortical semantics.

We project the predicted voxel-wise attention weights across the entire test dataset into a three-dimensional manifold using UMAP. The resulting embedding exhibits clear semantic clustering across higher visual cortex. This emergent organization mirrors known representational gradients in visual areas, demonstrating that our model internalizes not merely *how* to perform functional inversion, but *where* to find semantically relevant neural representations.

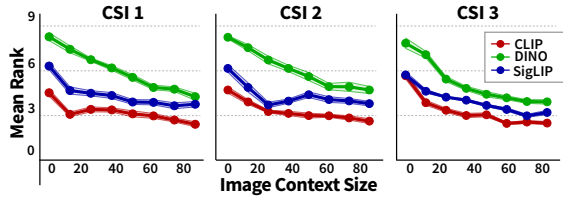


Figure 7. **Image-context scaling on BOLD5000.** Retrieval Mean rank (lower is better) across three unseen subjects as the number of in-context image-brain pairs increases.

4.5. New Scanner Adaptation on BOLD5000

We further assess cross-site generalization on BOLD5000, which differs substantially from NSD and thus provides a stringent test of new-scanner adaptation. Retrieval tasks are performed with 5-fold cross validation on BOLD5000 test images. Compared with NSD, BOLD5000 was acquired on a 3T scanner with different stimulus timing (slow event-related design with a 10 s inter-trial interval), a substantially different image set, a different voxel size (2 mm isotropic), and a different subject pool. Despite these shifts, BrainCoDec achieves remarkable results on strong retrieval performance and exhibits a similar contextual scaling trend (Figure 7, Figure 8). Results are consistent across held-out subjects and across image-encoder backbones (Table 2).

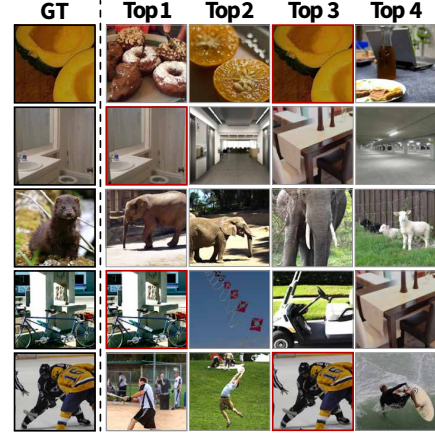


Figure 8. **Top-4 image retrieval on BOLD5000.** We visualize the retrieval result on a **new-scanner unseen subject** (CS11) using 20 images as context. The right columns display the Top-4 retrieved images. Red boxes indicate correct hits. Our model can generalize to novel datasets and scanning parameters without training.

Our model clearly could transfer its pretrained knowledge to new scanners, which is valuable in practical applications where retraining new models for new subjects is resource-intensive and time-consuming.

Table 2. **Quantitative results of BOLD5000.** We directly test BrainCoDec using just 20 images as in-context and a different 20 images as the test set on **three unseen subjects** from BOLD5000. Chance Top-1 Acc is 5%. All metrics are averaged across all folds of all three unseen subjects.

Backbones	Top-1 Acc.	Top-5 Acc.	Mean Rank	Cosine Sim.
CLIP	31.45±12.80%	81.67±9.42%	3.49±0.76	0.72±0.02
DINOv2	13.99±5.83%	53.33±6.74%	6.78±0.87	0.08±0.01
SigLIP	23.67±8.05%	73.41±8.25%	4.47±0.93	0.66±0.01

5. Conclusion

We present a foundation framework for fMRI decoding that generalizes across subjects, scanners, and acquisition protocols without any fine-tuning. By meta-learning how to invert visual encoding functions and performing hierarchical in-context inference across stimuli and voxels, BrainCoDec achieves substantial gains in data efficiency, interpretability, and cross-subject performance over strong baselines. Beyond decoding, our approach offers a principled computational lens on population-level cortical organization and demonstrates how learned functional inversion can scale across heterogeneous neural datasets. Looking forward, the same strategy can be extended to EEG, MEG, and other modalities, opening a pathway toward a universal, training-free neural decoding model for cognitive science, machine perception, and real-world BCIs.

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